

KOICHI ITO

Singapore

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Research Interests

A PhD candidate at the National University of Singapore and consultant at the World Bank Group, interested in researching travel behavior, human perception, and equity with emerging spatial data sources, such as street-view imagery, and machine/deep learning techniques. Creator of ZenSVI, a Python package with over 100,000 downloads, designed for analyzing street-view imagery data.

Education

The National University of Singapore

Aug 2022 – Present

PhD in Urban Analytics

Singapore

- Ongoing projects supervised by Dr. Filip Biljecki:

1. Sequential and agent-based modeling of the effects of cycling interventions on intention to cycle

The National University of Singapore

Aug 2019 – May 2021

Master of Urban Planning

Singapore

- 4.53/5.00 cumulative GPA (Highest Distinction)

Soka University of America

Aug 2015 – May 2019

Bachelor of Arts, Liberal Arts with Concentration in Environmental Studies

Aliso Viejo, USA

- 3.95 /4.00 cumulative GPA (summa cum laude)

Council on International Educational Exchange

Aug 2017 – Dec 2017

Study Abroad Program

Dakar, Senegal

Visiting Researcher Experience

Delft University of Technology

May 2025 – July 2025

Visiting Researcher, CityAI Lab

Delft, the Netherlands

The University of Texas at Austin

Feb 2025 – Apr 2025

Visiting Researcher, GISense Lab

Austin, USA

The University of Tokyo

Jan 2025

Visiting Researcher, the Urban Transportation Research Unit

Tokyo, Japan

Honors

- Singapore International Graduate Award (SINGA) Scholarship Aug 2022 – Present
- Japan Student Services Organization Scholarship (US\$40,000) Aug 2020 – May 2021
- Soka University of America Makiguchi Scholarship (US\$10,000) May 2019
- Soka University of America Dean's List (GPA is 3.7 or higher) Dec 2015 – May 2019
- Soka University of America Merit Scholarship (US\$20,000) Aug 2015 – May 2019

Publications

11. Ito, K., et al. (2025). *ZenSVI: An open-source software for the integrated acquisition, processing and analysis of street view imagery towards scalable urban science*. *Computers, Environment and Urban Systems*, 119, 102283. <https://doi.org/10.1016/j.compenvurbsys.2025.102283>
10. Gu, Y., et al. (Ito, K.) (2025). *Designing effective image-based surveys for urban visual perception*. *Landscape and Urban Planning*, 260, 105368. <https://doi.org/10.1016/j.landurbplan.2025.105368>
9. Ito, K., Bansal, P., & Biljecki, F. (2024). *Examining the causal impacts of the built environment on active transportation using time-series street view imagery*. *Transportation Research Part A: Policy and Practice*, 190, 104286. <https://doi.org/10.1016/j.tra.2024.104286>
8. Ito, K., Quintana, M., Han, X., Zimmermann, R., & Biljecki, F. (2024). *Translating street view imagery to correct perspectives to enhance bikeability and walkability studies*. *International Journal of Geographical Information Science*, 38, 2514. <https://doi.org/10.1080/13658816.2024.2391969>

7. **Ito, K.**, Kang, Y., Zhang, Y., Zhang F., & Biljecki, F. (2024). *Understanding Urban Perception with Visual Data: A Systematic Review*. *Cities*, 152, 105169. <https://doi.org/10.1016/j.cities.2024.105169>
6. Fujiwara, K., **Ito, K.**, Ignatius, M., & Biljecki, F. (2024). *A panorama-based technique to estimate sky view factor and solar irradiance considering transmittance of tree canopies*. *Building and Environment*, 266, 112071. <https://doi.org/10.1016/j.buildenv.2024.112071>
5. Hou, Y., et al. (**Ito, K.**) (2024). *Global Streetscapes – A comprehensive dataset of millions of street-level images over 700 cities for urban science and analytics*. *ISPRS Journal of Photogrammetry and Remote Sensing*, 215. <https://doi.org/10.1016/j.isprsjprs.2024.06.023>
4. Wang, S., et al. (**Ito, K.**) (2024). *Mapping the landscape and roadmap of geospatial artificial intelligence (GeoAI) in quantitative human geography: An extensive systematic review*. *International Journal of Applied Earth Observations and Geoinformation*, 128, 103734. <https://doi.org/10.1016/j.jag.2024.103734>
3. Wang, Z., **Ito, K.**, & Biljecki, F. (2023). *Assessing the equity and evolution of urban visual perceptual quality with time series street view imagery*. *Cities*, 145, 104704. <https://doi.org/10.1016/j.cities.2023.104704>
2. **Ito, K.**, & Biljecki, F. (2021). *Assessing bikeability with street view imagery and computer vision*. *Transportation Research Part C: Emerging Technologies*, 132, 103371. <https://doi.org/10.1016/j.trc.2021.103371>
1. Biljecki, F., & **Ito, K.** (2021). *Street view imagery in urban analytics and GIS: A review*. *Landscape and Urban Planning*, 215, 104217. <https://doi.org/10.1016/j.landurbplan.2021.104217>

Academic Services

Peer Reviews

Cities (5 times), Landscape and Urban Planning (4 times), Urban Informatics (2 times), International Journal of Geographical Information Science (2 times), Environment and Planning B: Urban Analytics and City Science (2 times), Transportation Research Part A: Policy and Practice (1 time), Transport Geography (1 time), Health and Place (1 time), Patterns (1 time), ISPRS Journal of Photogrammetry and Remote Sensing (1 time), Scientific Data (1 time), Philosophical Transactions A (1 time), Computational Urban Science (1 time), Journal of Geographical Systems (1 time), International Journal of Applied Earth Observation and Geoinformation (1 time), Computers, Environment and Urban Systems (1 time)

Services at Organizations

- **International Forum on Urbanism (IFoU):** Young Ambassador
- 2024 – Present**

Invited Talks and Poster Presentations

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| 10. “Perceived Traffic Safety’s Impact on Cycling Route Choice”
Gave a guest lecture at TU Delft Faculty of Technology, Policy and Management | July 2025
<i>Delft, the Netherlands</i> |
| 9. “Seeing cities with human-centric visual AI”
Gave an invited talk at the Ghent University | May 2025
<i>Ghent, Belgium</i> |
| 8. “Programming for GIS”
Gave a guest lecture for GRG 356T Cartography And Maps at the University of Texas at Austin | March 2025
<i>Austin, USA</i> |
| 7. “Understanding Human Perceptions and Travel Behavior with Emerging Technologies”
Gave invited talks for Sekimoto Lab and Urban Transportation Research Unit at the University of Tokyo | Jan 2025
<i>Tokyo, Japan</i> |
| 6. “Examining the causal impacts of the built environment on cycling activities using time-series street view imagery”
Chaired a session and presented at the IATBR Conference | Jul 2024
<i>Vienna, Austria</i> |
| 5. “Advancing urban modeling with emerging geospatial datasets and AI technologies”
Gave an invited talk at the Global Space and Technology Convention 2024 Conference hosted by Singapore Land Authority. | Feb 2024
<i>Singapore</i> |
| 4. “Evaluating active mobility and urban perception with visual data”
Gave a guest lecture talk at a GIS course at the National University of Singapore. | Oct 2023
<i>Singapore</i> |
| 3. “Sidewalk the Talk: Translating street view imagery to correct perspectives to enhance bikeability and walkability studies”
Presented a poster at the Urban Solutions and Sustainability R&D Congress. | Oct 2023
<i>Singapore</i> |

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| 2. “Street View Imagery: Have we answered all the questions with it? What’s left to do?”
Chaired a session at Spatial Data Science Symposium 2023. | Sep 2023
<i>Online</i> |
| 1. “Application of GIS in research and international institution.”
Presented my research at Soka University of America, Advanced GIS Course. | Oct 2021
<i>Online</i> |

Work Experience

The World Bank Group — Poverty and Equity Global Practice	Apr 2021 – Jul 2022
<i>Consultant</i>	<i>Remote</i>

- Developed and maintained interactive data dashboards using Shiny and Golem frameworks to facilitate spatial data visualization and analysis.
- Implemented spatial lag models in R to analyze factors correlated with poverty and stunting.
- Led the collection, cleaning, and mapping of various spatial data sets to predict poverty rates in developing countries with XGBoost in Python.
- Conducted a capacity-building program for Tanzania’s National Bureau of Statistics, imparting skills and knowledge on spatial analysis using R.

Johnson Controls	Nov 2020 – May 2021
<i>Data Analytics Intern</i>	<i>Singapore</i>

- Constructed a data pipeline of PM2.5 data, weather data, and traffic data in Singapore on Azure Synapse for a real-time Power BI dashboard.
- Built a time-series model to predict PM 2.5 levels anywhere in Singapore with Python.

The World Bank Group — Tokyo Development Learning Center	Apr 2020 – Jan 2021
<i>Urban Development Research Intern</i>	<i>Remote</i>

- Compiled comprehensive data set of over 4,000 Tokyo-based startups’ information, such as addresses, investors, and accelerators, using Python.
- Visualized financial performance of light rail transport system in Toyama prefecture between 2002 and 2019.

ENDA LEAD Afrique Francophone	Aug 2017 – Dec 2017
<i>Research Intern</i>	<i>Dakar, Senegal</i>

- Authored a report on accessibility to potable water in 500 villages in the Saint-Louis region in Senegal.

Teaching Experience

CDE2212 AI for Design	Aug 2024 – Dec 2024
<i>Teaching Assistant</i>	<i>The National University of Singapore, Department of Architecture</i>

- Assisted in a comprehensive course on AI applications in design (generative models), providing in-person and online support during weekly lectures and tutorials, conducting lab sessions, and grading.

DEP5111 Planning Technologies	Jan 2024 – May 2024
<i>Teaching Assistant</i>	<i>The National University of Singapore, Department of Architecture</i>

- Assisted Master of Urban Planning students in mastering R Data Science through hands-on coding support, troubleshooting, and grading, enhancing their data analysis and visualization skills.

AR5805 Advanced Architecture Studio	Aug 2023 – Dec 2023
<i>Teaching Assistant</i>	<i>The National University of Singapore, Department of Architecture</i>

- Assisted the lead professor with logistical and administrative tasks, including the preparation of course documents and aiding in the smooth operation of lectures.

Supervising

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| • Shu Wang: Master of Applied GIS, the National University of Singapore | Apr 2024 – Jul 2024 |
| • Heyang Hua: Master of Urban Planning, the National University of Singapore | Jan 2024 – May 2024 |
| • Zeyu Wang: Master of Urban Planning, the National University of Singapore | Aug 2022 – May 2023 |

Technical Skills

Languages: Japanese (native), English (fluent), French (limited working proficiency)
Computer Skills: Python, R, SQL, Spark, ArcGIS, QGIS, Google Earth Engine, Azure, AWS, Power BI

References

- **Filip Biljecki**
*Presidential Young
(Assistant) Professor*
College of Design and
Engineering, The National
University of Singapore
- **Prateek Bansal**
*Presidential Young
(Assistant) Professor*
College of Design and
Engineering, The National
University of Singapore
- **Zhang Ye**
Associate Professor
School of Architecture,
Tsinghua University
- **Takaaki Masaki**
Senior Economist
The Asian Development
Bank